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Project Name: **GLOBAL FOOD PRICE
ANALYSIS**

12-June-2026

Executive Summary

- The Global Food Price Analysis project runs from 2010 to 2017. It covers 55 countries with exchange rate data. All food prices are converted into US dollars. Data comes from Kaggle and the World Bank. Excel and Power BI are used to clean and visualize the dataset. The dataset holds more than 740,000 rows. A star schema is applied with fact and dimension tables. Dashboards are built for markets, commodities, volatility, and FX trends. The project serves governments, international organizations, analysts, and supply chain firms. One challenge is aligning monthly food prices with yearly exchange rate data. Another challenge is formula errors. Recommendations include adding newer data, monthly FX rates, and predictive analytics.

Background

- Food prices matter because they affect families, economies, and food security. Leaders and aid groups watch them to see trends and plan actions.
- Comparing prices across countries is tricky since currencies differ and exchange rates keep changing. To solve this, the project converts all prices into US dollars, making comparisons fair and clear.
- Power BI is used to clean the data and show it in charts and dashboards. This makes patterns and changes easy to see, helping people make better decisions.

Stakeholders & BI Requirements

Government Ministry (Agriculture / Trade / Finance)

Requirements:

- Monitor food price changes across countries
- Identify countries with high or unstable food prices
- Track impact of exchange rates on food prices
- Support policy and pricing decisions

Market Analysts

Requirements:

- Analyze commodity price behavior across markets
- Identify trends, volatility, and price patterns
- Compare pricing across countries and time

International Organizations (FAO / WFP / World Bank)

Requirements:

- Detect food insecurity risk through price levels and volatility
- Identify regions with unstable or high-cost food markets
- Support global food aid and intervention planning

Supply Chain Firms

Requirements:

- Identify price trends for planning procurement
- Understand commodity stability for risk reduction
- Compare markets for sourcing decisions

Analytical Questions

1. What is the overall structure of the dataset in terms of countries, markets, commodities, and average food prices (USD)?
2. How do food prices vary across countries and continents?
3. Which countries have the highest average food prices?
4. How are commodities distributed in the dataset?
5. How do food prices change over time (2010–2017)?
6. What is the relationship between exchange rates (FX) and food prices?
7. Which commodities show the highest price increases or decreases over time?
8. Which commodities are the most and least volatile?
9. What is the relationship between commodity price level and price volatility?
10. Which commodities are most frequently occurring in the dataset?

Project Workflow

1

Dataset

- ❑ Global Food Dataset retrieved from an Excel file
- ❑ Exchange rates (USD FX) pulled from the World Bank Api
- ❑ Dataset represents:
 - ❖ Commodities: Food
 - ❖ Time Period: 2010 – 2017
 - ❖ Countries: With available FX data for USD conversion

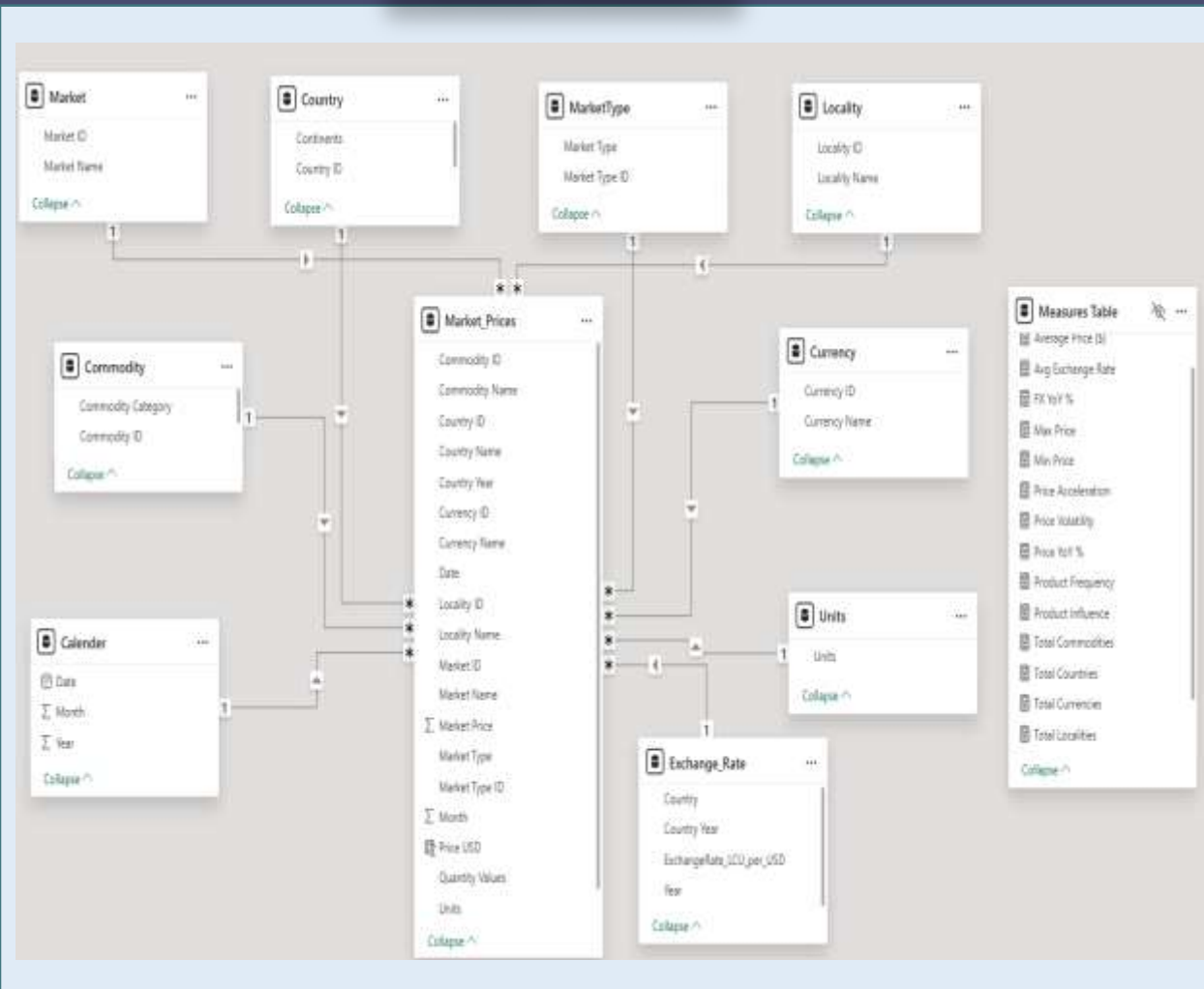
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Data Collection & Preprocessing (ETL)

- ❑ Data imported into Power BI (Power Query)
- ❑ Cleaning and transformation performed:
 - ❖ Standardizing country names
 - ❖ Cleaning market type values
 - ❖ Structuring continent grouping
- ❑ Preparation of dataset for modelling

3

Data Model



4

Report View

Market Overview Dashboard

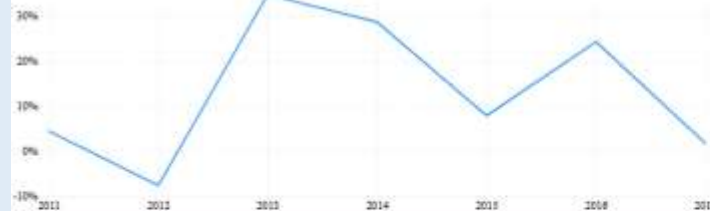


Report View

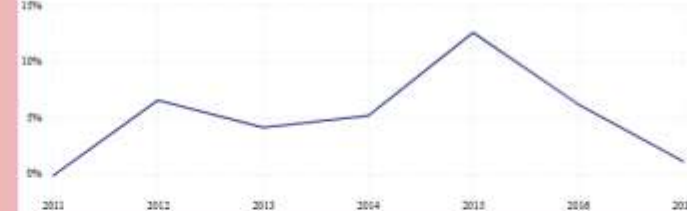
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Price Analysis Dashboard

Year-on-Year Commodity Price Changes (USD)



Year-on-Year FX Change



Year-on-Year FX Change

Year	FX YoY %
2011	-0.18%
2012	6.52%
2013	4.09%
2014	5.14%
2015	12.54%
2016	6.18%
2017	1.07%

Market Type

- ☐ Farm Gate
- ☐ Producer
- ☐ Retail
- ☐ Wholesale

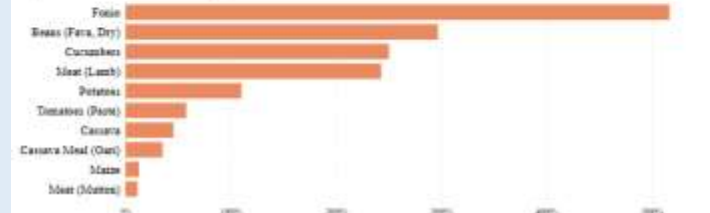
Continents

- ☐ Africa
- ☐ Asia
- ☐ Europe
- ☐ North America
- ☐ Other
- ☐ South America

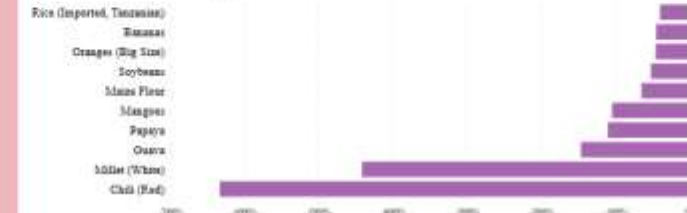
Year-on-Year Price Change

Year	Price YoY %
2011	4.43%
2012	-7.68%
2013	34.80%
2014	28.71%
2015	7.90%
2016	24.32%
2017	1.76%

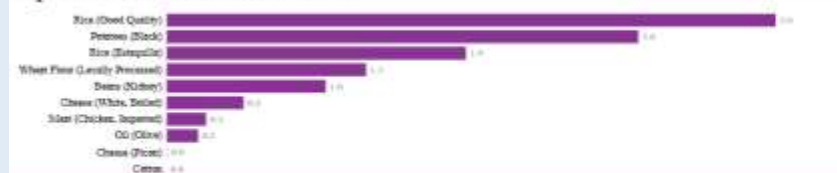
Top 10 Price Accelerating Commodities



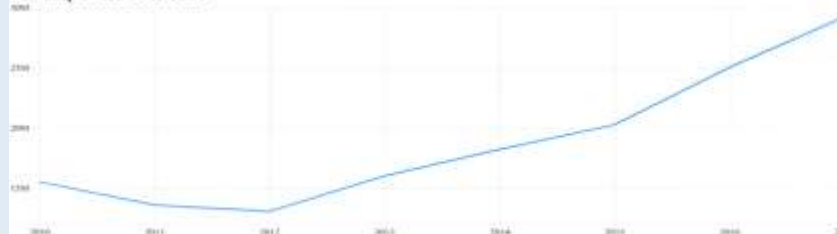
Top 10 Price Decelerating Commodities



Top 10 Least Volatile Commodities



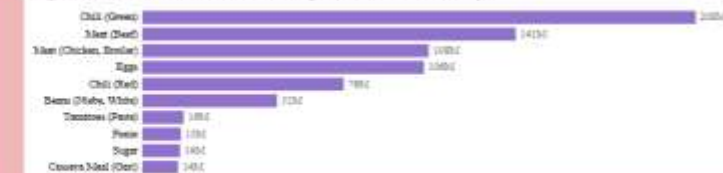
Volatility Trend Over Time



Price vs Volatility of Commodities (USD)



Top 10 Commodities with the Highest Price Volatility



Top 10 Most Volatile Commodities

Commodity Name	Price Volatility
Chili (Green)	208,367,648.87
Meat (Beef)	140,843,948.87
Meat (Chicken, Broiler)	107,816,660.75
Eggs	106,054,021.21
Chili (Red)	75,827,856.80
Beans (Nieto, White)	50,774,734.74
Tomatoes (Paste)	13,398,818.74
Fonio	14,574,441.45
Sugar	14,229,943.70
Cassava Meal (Gari)	13,520,947.46

Market Type

- ☐ Farm Gate
- ☐ Producer
- ☐ Retail
- ☐ Wholesale

Continents

- ☐ Africa
- ☐ Asia
- ☐ Europe
- ☐ North America
- ☐ Other
- ☐ South America

Top 10 Commodities by Frequency

Commodity Name	Product Frequency
Sorghum	21062
Maize	20312
Rice (Imported)	18957
Millet	18683
Rice (Local)	15057
Rice	14372
Maize (White)	13187
Sugar	12404
Wheat Flour	12394
Wheat	9763
Beans (Dry)	9540
Groundnuts (Shelled)	8936
Tomatoes	7644
Onions	5871
Potatoes	5518

Top 10 Least Volatile Commodities

Commodity Name	Price Volatility
Rice (Good Quality)	3.89
Potatoes (Black)	3.01
Rice (Istapuilla)	1.91
Wheat Flour (Locally Processed)	1.27
Beans (Kidney)	1.01
Cheese (White, Boiled)	0.49
Meat (Chicken, Imported)	0.23
Oil (Olive)	0.20
Cheese (Picon)	0.00
Cotton	0.00

Product Analysis Dashboard

User Journey

START

Market Overview

- KPI Summary
- Country & Continent Analysis
- Market & Commodity Overview

Price Analysis

- YoY Price Trends
- FX Trends
- Accelerating & Decelerating Products

Product Analysis

- Volatility Analysis
- Price vs Volatility
- Product Frequency

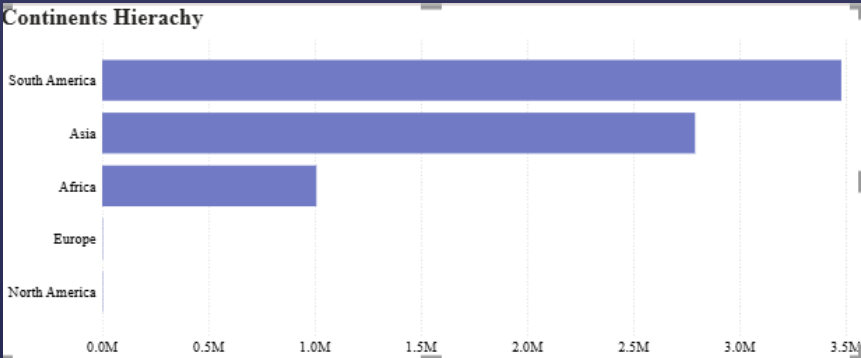
Period: 2010 – 2017
Commodity: Food
Countries: Those with FX Data

Insights

Overall Market

No. of Countries	No. of Comm. Categories	No. of Commodities	No. of Markets	Average Price (\$)
55	88	218	1234	1,548,369

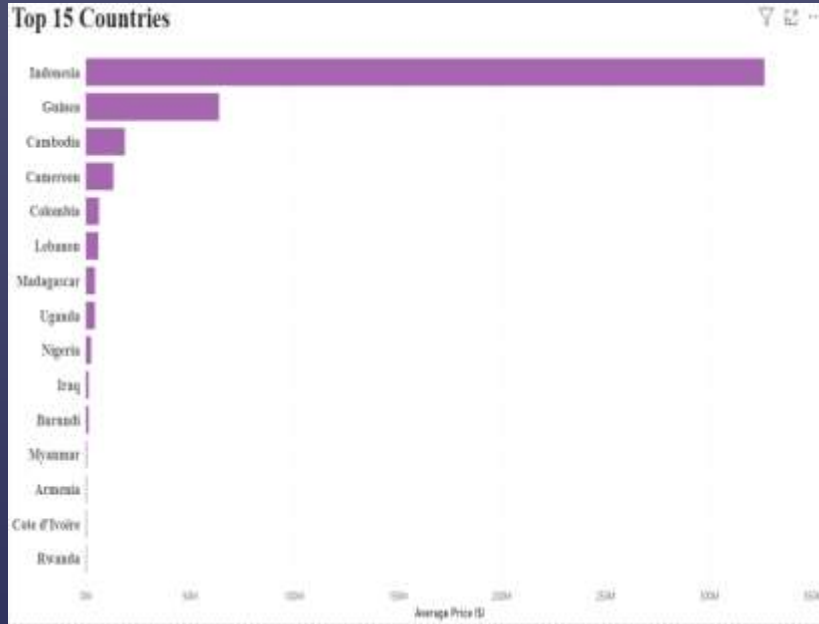
Continents



Continents	Total Commodities	Average Price (\$)
South America	59	3,478,444
Asia	106	2,790,440
Africa	124	1,008,201
Europe	17	689
North America	8	245

- **South America** tops the list at **\$3.48M** with only **59** commodities. High prices here aren't about variety.
 - **Asia** follows at **\$2.79M** across **106** commodities. Still fewer items, but strong price levels.
 - **Africa** averages **\$1.01M** with the widest range, **124** commodities. More varieties, yet lower prices.
 - **Europe** shows **\$689** from **17** commodities. Too little data to reflect real markets.
 - **North America** records **\$245** with just **8** commodities. Also not meaningful..
- **South America and Asia** currencies' are stronger and more stable against the US Dollar. When prices are converted, they keep more of their value. That's why average food prices look higher in these regions.
 - **African** currencies are weaker against the Dollar. This makes converted prices look lower, even though the real cost for local people may be heavier than it appears.

Countries



➤ Indonesia

Indonesia is the biggest outlier. Its average food price is **\$326.58M**, far above everyone else. This points to either very high price levels or unusual values that need closer review.

➤ Gap to Guinea

Second place is Guinea at **\$64.10M**. That's a drop of more than **\$262M** compared to Indonesia. The gap between the top two is huge.

➤ Top five

Indonesia, Guinea, Cambodia, Cameroon, and Colombia dominate the ranking. Their prices are much higher than the rest.

➤ Rapid decline

After the leaders, prices fall fast. From Indonesia's \$326.58M down to Rwanda's \$377,059 in 15th place, the difference is steep and uneven.

➤ Disparity

Food prices are not evenly spread. They vary a lot from one country to another.

➤ Middle ranks

Countries ranked around 10th to 15th are closer in value. The biggest differences are concentrated at the very top.

Commodities

Top 10 Commodities

Commodity Name Average Price (\$)

Chili (Green)	385,152,460
Meat (Chicken, Broiler)	375,237,722
Fish	222,663,828
Eggs	97,296,821
Beans (Niebe, White)	70,526,042
Meat (Pork, With Fat)	56,446,108
Meat (Beef)	26,233,714
Oil (Vegetable)	18,807,321
Milk (Powder)	18,390,870
Peas (Yellow)	15,850,979

- Significant price drop between Fish (**\$222.66 M**) and Eggs (**97.29 M**)

➤ Lower Price Group

Vegetable oil, powdered milk, and yellow peas all record less than **\$20M**.

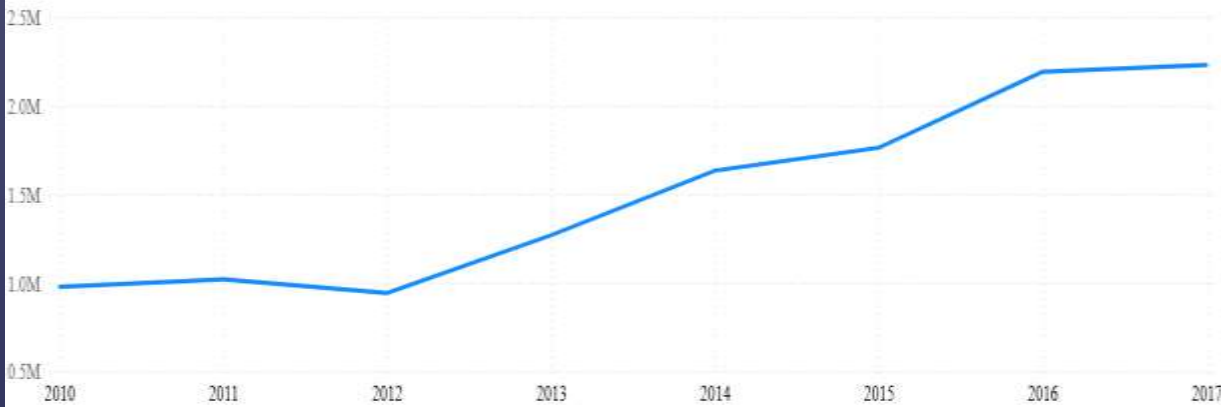
- Prices are highly skewed.
- A few commodities show extremely high averages, while most are much lower.

Time Trends

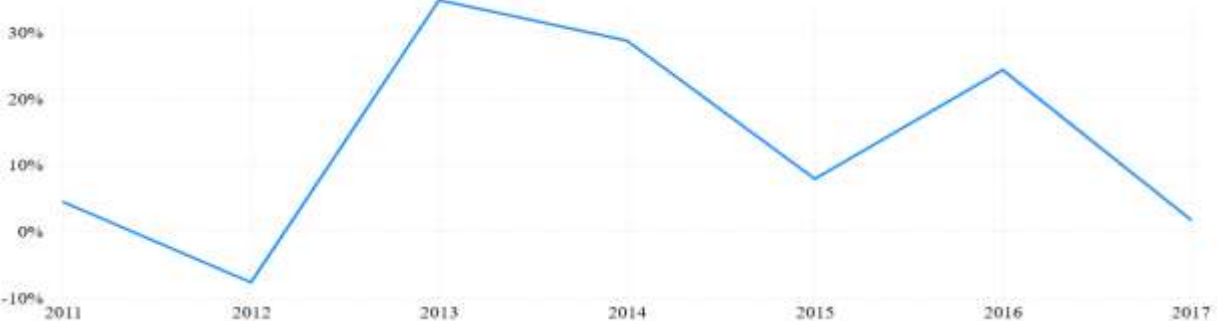
Drilled to the lowest level of data (Quarter)

Highest level data (Year)

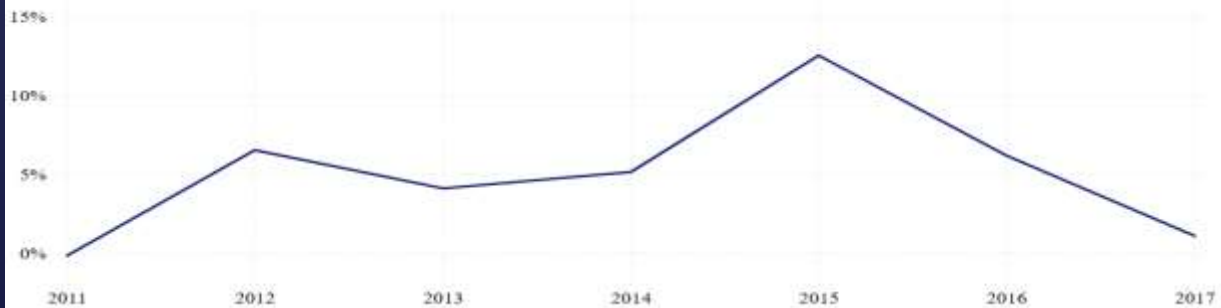
Price Trend



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- When YoY FX rates change, the prices also change.

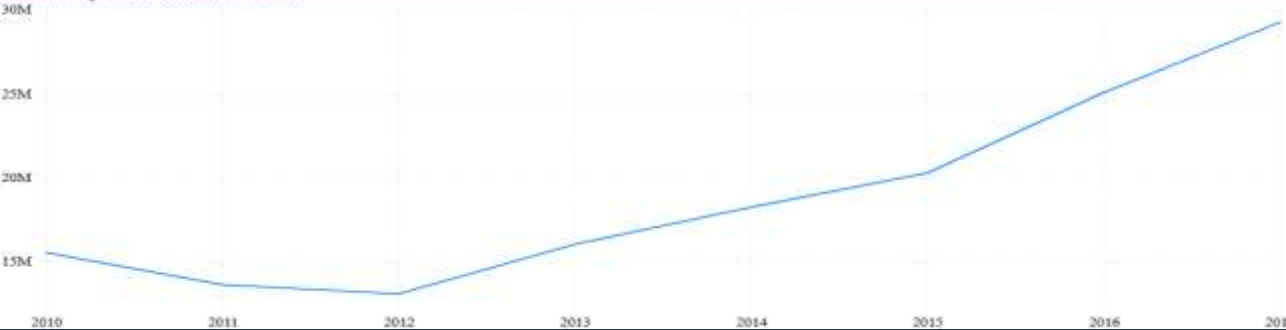
- Increase in FX rate, increases the Price

Volatility

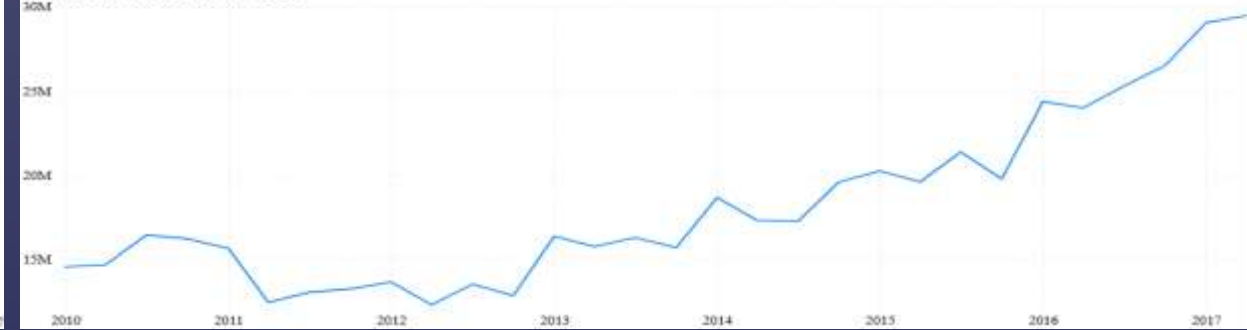
- High volatility:** high uncertainty/risky commodities.

- Low volatility:** stable commodities

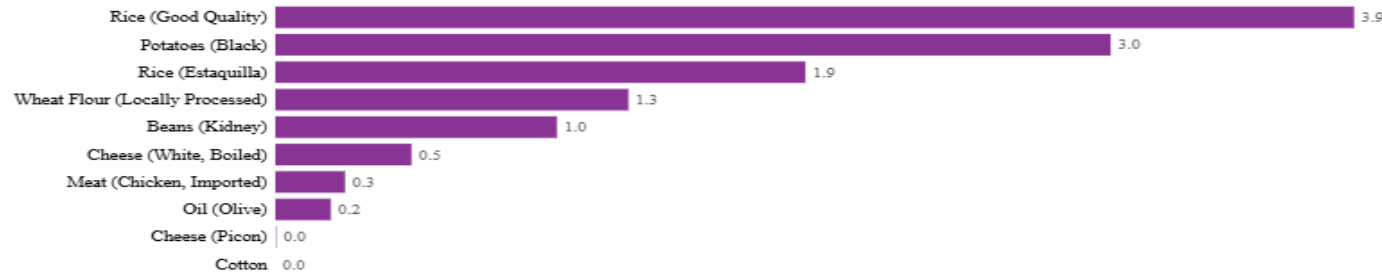
Volatility Trend Over Time



Volatility Trend Over Time



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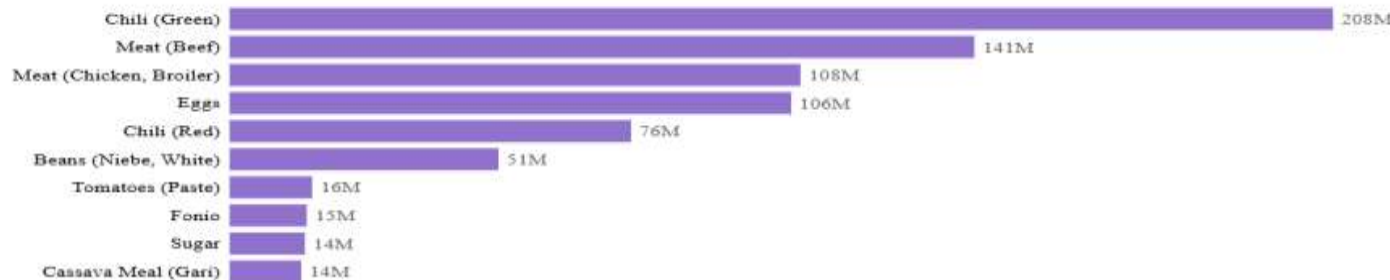


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Oil (Olive)	0.20
Cheese (Picon)	0.00
Cotton	0.00

- 2015 Qtr4 – 2017 Qtr2 price volatility has been increasing substantially

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Top 10 Least Volatile Commodities



• Volatility

From **2015 Q4** to **2017 Q2**, prices moved up and down a lot more. The market looked unstable, with bigger swings, even though overall prices didn't really rise.

• Exchange rates

In the same period, Year-over-Year exchange rate growth kept falling. By **2017 Q2**, currencies were putting less pressure on prices than they did in **2015 Q4**.

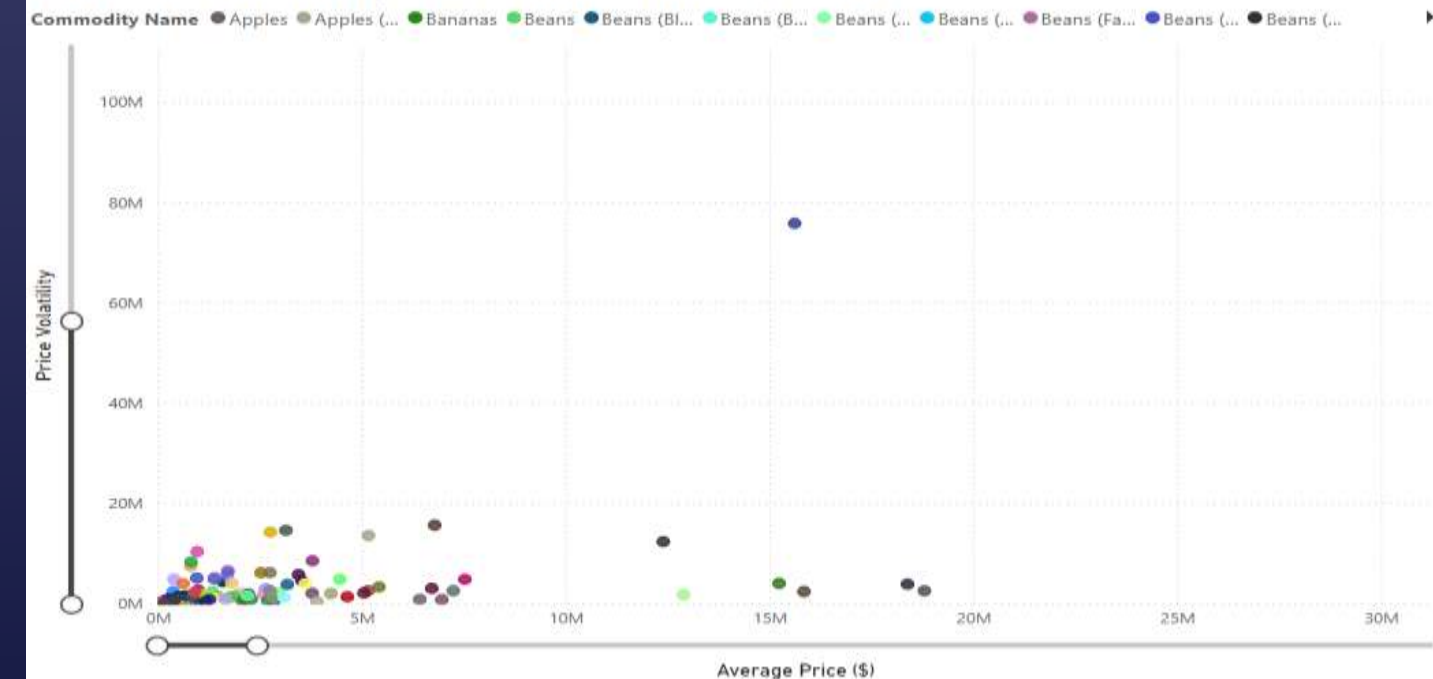
• Price growth

Between **2016 Q1** and **2017 Q4**, the speed of price increases slowed. Prices still shifted, but the upward push lost strength.

• Stability vs. volatility

From **2016 Q1** to **2017 Q4**, average prices stayed steady. The market had short-term jumps and drops, but no lasting rise in overall levels.

Price vs Volatility of Commodities (USD)



• Clustered

Most commodities sit close together. On average, they're around **USD 3M**. Their volatility is under **10%**, which means prices are moderate and don't swing too much.

• Outliers

A few commodities break away from this calm picture. They show much higher or more unstable prices, clearly standing out from the rest. Overall, the market is steady and mid-range, though there're a lot of commodities that stand out.

Technical Specifications

Data Source

Dataset	Source	Format	Size
Global Food Prices	Kaggle	.csv	740,000+ rows
Exchange Rate (LCU per USD)	World Bank	.xlsx/.csv	Annual, 74 countries

Environment

Item	Detail
Tool	Power BI Desktop + Excel
Refresh	Static — one-time load
Currency	USD

Data Model

Design:

- Star Schema

Fact Table:

- Market_Prices

Dimension Tables:

- Calendar
- Country
- Commodity
- Market
- MarketType
- Locality
- Currency
- Units

Reference Table

- Exchange_Rate

ETL Summary

Step	Detail
Tool	Excel + Power BI Power Query
Filters Applied	✓ 2010–2017 ✓ Food only ✓ 55 countries with FX data
Date Column	Created via <code>#date([Year], [Month], 1)</code>
Dimension Tables	Extracted from fact table into separate tables
Null Handling	Rows with missing prices excluded

DAX Formulas

- Average Price (\$)
- Price YoY %
- FX YoY %
- Price Volatility
- Price Acceleration
- Total Countries
- Total Commodities
- Total Markets
- Product Frequency

Assumption

- ❑ **Time Period:** Data covers 2010 – 2017
- ❑ **Commodities:** Focus is on food items only
- ❑ **Currency:** All prices converted to US dollars for fair comparison.
- ❑ **Exchange rates:** World Bank gives rates by year. For monthly prices, the yearly average was used.
- ❑ **Volatility:** Measured by standard deviation, showing stable vs unstable foods.

User Accessibility Design

- ❑ **Interactive slicers:** All dashboards have slicers for market type and continent. This lets one focus on the parts that matter most to them.
- ❑ **Visual variety:** Data is shown in many ways. There are bars, lines, pies, scatter plots, and tables. This gives users options depending on how they like to see information.
- ❑ **Consistent filters** All pages use the same filters: years 2010 to 2017, food items only, and 55 countries. This keeps everything aligned and easy to follow.

Privacy Considerations

- ❑ **Public data:** The project only uses open sources: Kaggle for food prices and the World Bank for exchange rates. No personal details are included.
- ❑ **Intended use** The report is for policy, research, and market analysis. It's not for resale without credit to the sources.

Challenges

- Combining the 2 data sources was difficult
- Food prices were monthly. Exchange rates were yearly. This mismatch made alignment hard
- Incorrect relationships between Market_Prices (Fact table), Calendar, and Exchange Rates, resulted to some DAX formula producing incorrect results e.g. Price YoY %, FX YoY %
- Visualizing FX YoY % and Price YoY % in a single line & column visual proved difficult
- The original dataset contained local currencies for 55+ countries. Analyzing the data with these local currencies gave inaccurate results.

THANK YOU!